

# Лекция 5

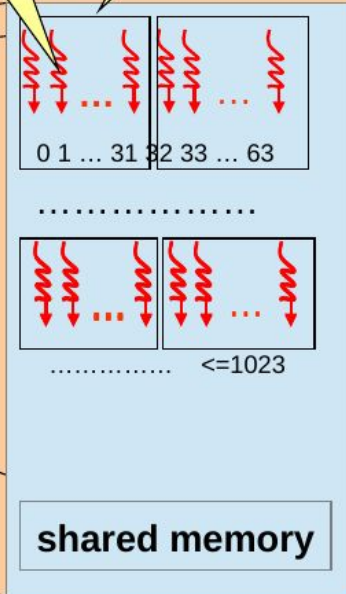
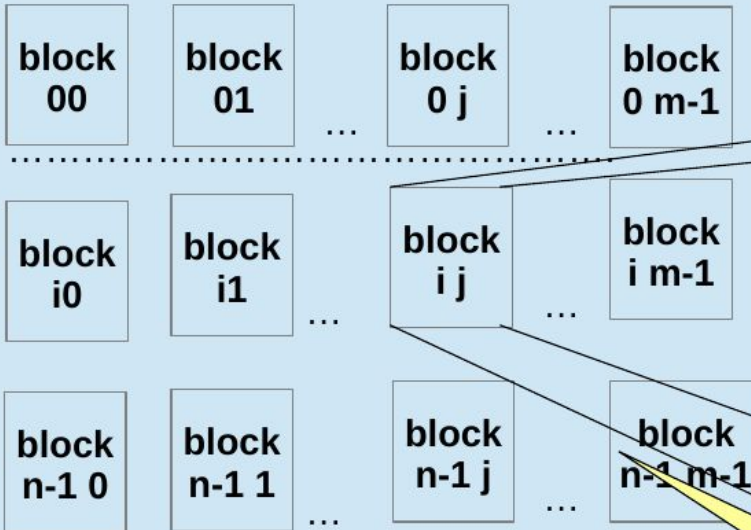
- Объединение нитей в блоки и варпы.
- Оптимальная конфигурация нитей.
- Иерархия памяти.

**device**

grid

thread

warp



$n \times m \leq 65536$

**Global memory**

копирование device -> host

```
#include <stdio.h>
#include <stdlib.h>

global void gShowIdx() {
int idx = blockIdx.x * blockDim.x + threadIdx.x;
int warp_idx = threadIdx.x / warpSize;
int lane_idx = threadIdx.x % warpSize;
printf(" %5d\t%5d\t %2d\t%2d\n",
        idx, blockIdx.x, warp_idx, lane_idx);
}
```

```
int main(int argc, char** argv){
    if(argc < 3){
        fprintf(stderr,
            "USAGE: <prog> <threads_per_block> <num_of_blocks>\n");
        return -1;
    }

    int threads_per_block=atoi(argv[1]);
    int num_of_blocks=atoi(argv[2]);

    gShowIdx<<<num_of_blocks, threads_per_block>>>();
    cudaDeviceSynchronize();

    return 0;
}
```

```
./lab4a 64 2 | sort -g -k1,1 -k2,2  
-k4,4 > stat.txt
```

| <i>idx</i> | <i>blk</i> | <i>wrp</i> | <i>lane</i> |
|------------|------------|------------|-------------|
|------------|------------|------------|-------------|

|   |   |   |   |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
|---|---|---|---|

|   |   |   |   |
|---|---|---|---|
| 1 | 0 | 0 | 1 |
|---|---|---|---|

|   |   |   |   |
|---|---|---|---|
| 2 | 0 | 0 | 2 |
|---|---|---|---|

|    |   |   |    |
|----|---|---|----|
| 30 | 0 | 0 | 30 |
|----|---|---|----|

|    |   |   |    |
|----|---|---|----|
| 31 | 0 | 0 | 31 |
|----|---|---|----|

|    |   |   |   |
|----|---|---|---|
| 32 | 0 | 1 | 0 |
|----|---|---|---|

|    |   |   |   |
|----|---|---|---|
| 33 | 0 | 1 | 1 |
|----|---|---|---|

|    |   |   |   |
|----|---|---|---|
| 34 | 0 | 1 | 2 |
|----|---|---|---|

|    |   |   |    |
|----|---|---|----|
| 62 | 0 | 1 | 30 |
|----|---|---|----|

|    |   |   |    |
|----|---|---|----|
| 63 | 0 | 1 | 31 |
|----|---|---|----|

|    |   |   |   |
|----|---|---|---|
| 64 | 1 | 0 | 0 |
|----|---|---|---|

|    |   |   |   |
|----|---|---|---|
| 65 | 1 | 0 | 1 |
|----|---|---|---|

|    |   |   |    |
|----|---|---|----|
| 95 | 1 | 0 | 31 |
|----|---|---|----|

|    |   |   |   |
|----|---|---|---|
| 96 | 1 | 1 | 0 |
|----|---|---|---|

|    |   |   |   |
|----|---|---|---|
| 97 | 1 | 1 | 1 |
|----|---|---|---|

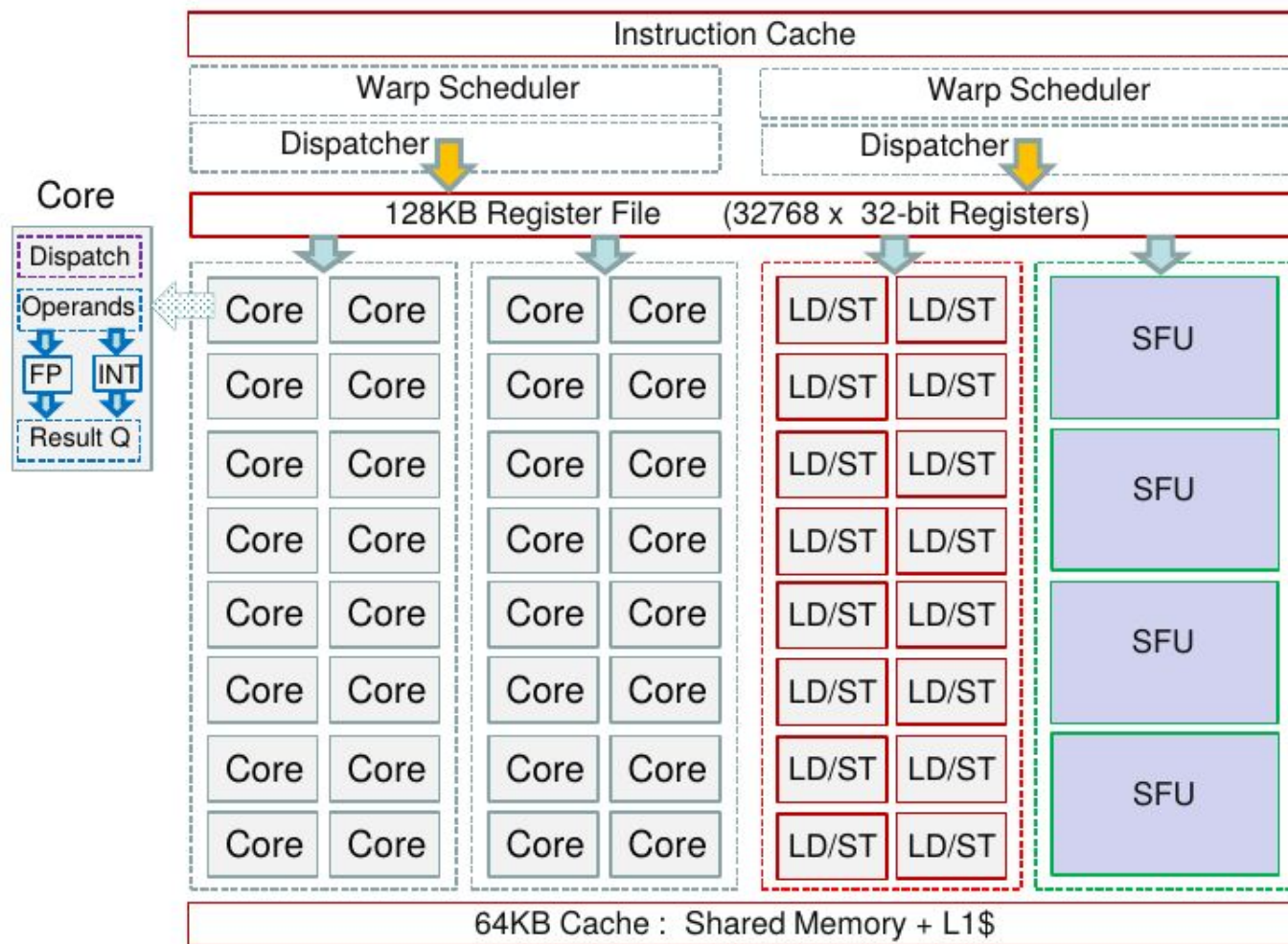
|     |   |   |    |
|-----|---|---|----|
| 127 | 1 | 1 | 31 |
|-----|---|---|----|

# Оптимальное количество нитей в блоках (сокращение латентности)

Преимущества  
перекрывания/многозадачности:



Кол-во варпов  $\% 32 == 0 \ \&\& \ \text{Кол-во варпов} / 32 > 1$   
Кол-во блоков  $\geq$  Кол-во мультипроцессоров



Detected 1 CUDA Capable device(s)

Device 0: "NVIDIA GeForce RTX 2060"

CUDA Driver Version / Runtime Version 12.0 / 11.1

CUDA Capability Major/Minor version number: 7.5

.....

(**30**) Multiprocessors, ( **64**) CUDA Cores/MP: 1920 CUDA Cores

.....

Memory Clock rate: 7001 Mhz

Memory Bus Width: 192-bit

L2 Cache Size: 3145728 bytes

.....

Total amount of constant memory: 65536 bytes

Total amount of shared memory per block: 49152 bytes

Total shared memory per multiprocessor: 65536 bytes

Total number of registers available per block: 65536

Warp size: 32

Maximum number of threads per multiprocessor: 1024

Maximum number of threads per block: 1024



```
Detected 2 CUDA Capable device(s)
Device 0: "GeForce GTX 1050"
  CUDA Driver Version / Runtime Version          9.1 / 9.1
  CUDA Capability Major/Minor version number:    6.1
.....
( 5) Multiprocessors, (128) CUDA Cores/MP:      640 CUDA Cores
.....
Memory Clock rate:                             3504 Mhz
Memory Bus Width:                              128-bit
L2 Cache Size:                                 1048576 bytes
.....
Total amount of constant memory:                65536 bytes
Total amount of shared memory per block:        49152 bytes
Total number of registers available per block:  65536
Warp size:                                       32
Maximum number of threads per multiprocessor:   2048
Maximum number of threads per block:            1024
.....
```



```
global void gSum(int* a, int *b){  
    int i=threadIdx.x+blockIdx.x*blockDim.x;  
    a[i]+=b[i];  
}
```

```
int N=1<<atoi(argv[1]);  
int num_threads=atoi(argv[2]);  
int num_blocks=N/num_threads;
```

```
gSum<<<num_blocks, num_threads>>>(a,b);  
cudaDeviceSynchronize();  
CUDA_CHECK_RETURN(cudaGetLastError());
```

```
/Lab4 # ncu --target-processes all -k gSum ./lab4 20 32
```

gSum(int \*, int \*), 2023-Feb-20 17:05:19, Context 1, Stream 7

Section: GPU Speed Of Light Throughput

-----  
**Duration**                      **usecond**                      **69.38**  
-----

Section: Occupancy

-----  
**Theoretical Occupancy**      %                      **50**  
**Achieved Occupancy**        %                      **33.63**  
..... . .

```
/Lab4 # ncu --target-processes all -k gSum ./lab4 20 64
```

gSum(int \*, int \*), 2023-Feb-20 17:05:19, Context 1, Stream 7

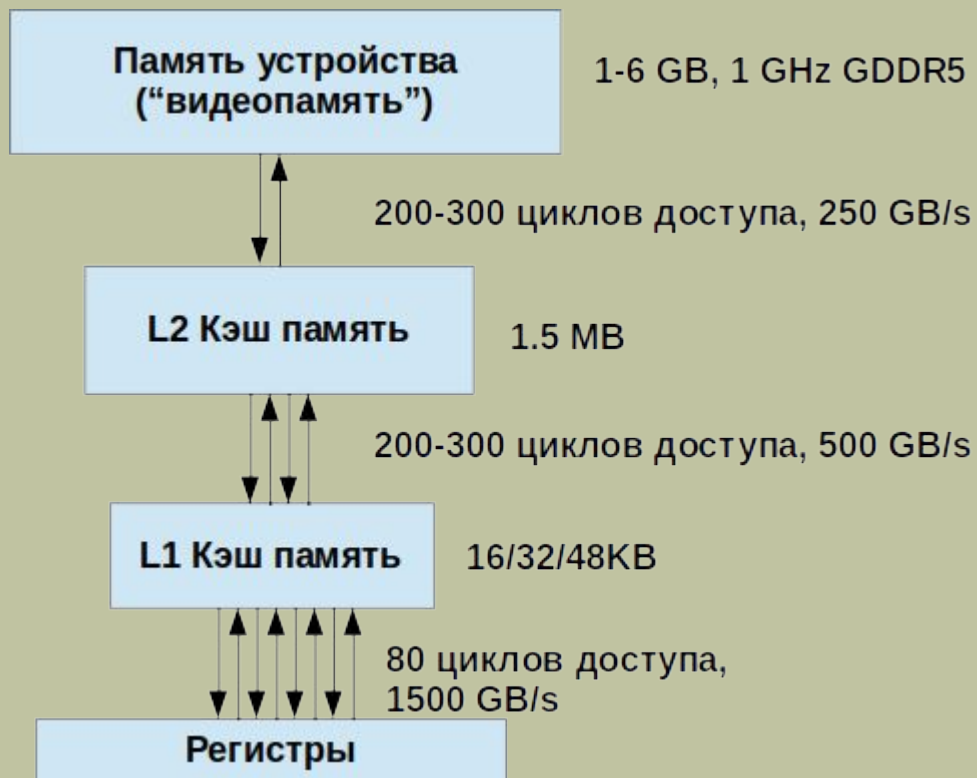
Section: GPU Speed Of Light Throughput

-----  
**Duration**                      **usecond**                      **42.78**  
-----

Section: Occupancy

-----  
**Theoretical Occupancy**      %                      **100**  
**Achieved Occupancy**        %                      **79.17**

..... . .



## Device

DRAM

*Local*

*Global*

*Constant*

*Texture*

To Host

GPU

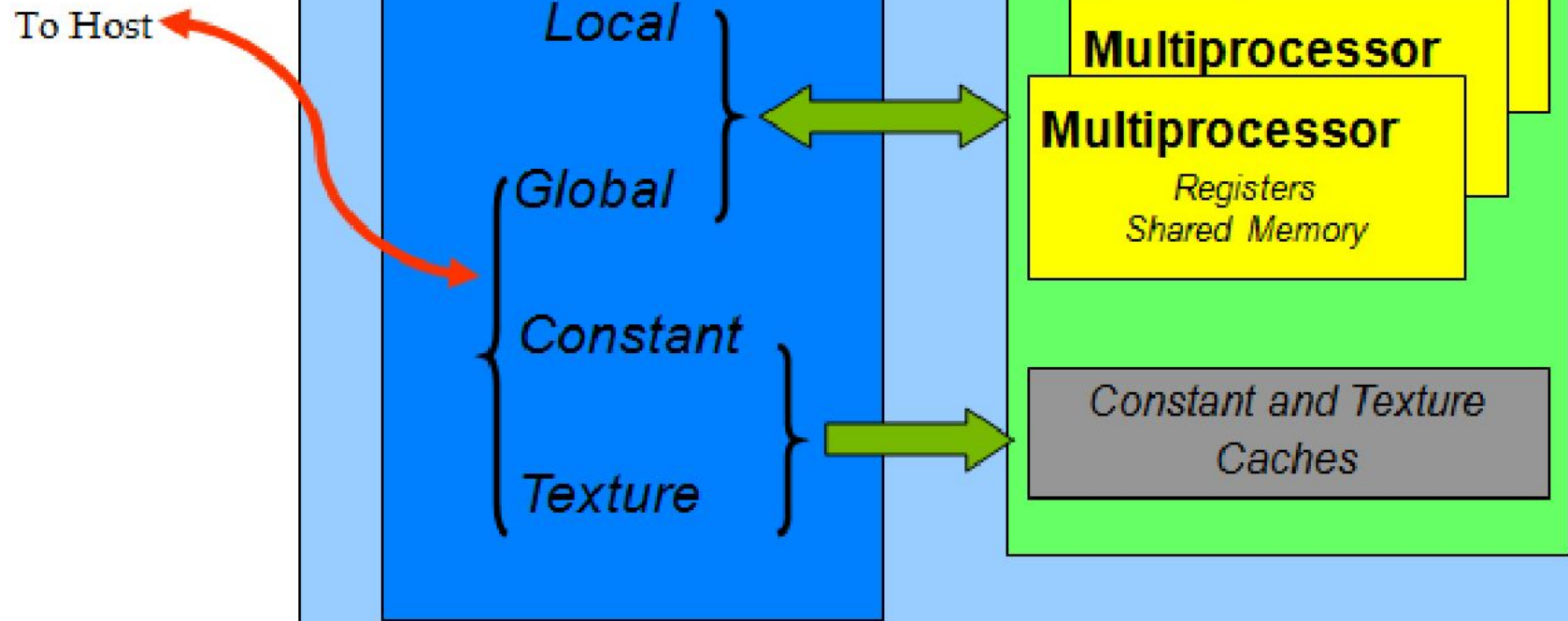
**Multiprocessor**

**Multiprocessor**

**Multiprocessor**

*Registers*  
*Shared Memory*

*Constant and Texture*  
*Caches*



```
#define DUMMY_LENGTH 102
```

```
__global__ void gInit(int* a, int* b){  
    int i=threadIdx.x+blockIdx.x*blockDim.x;  
    int dummy[DUMMY_LENGTH];  
    for(int j=0;j<DUMMY_LENGTH;j++)  
        dummy[j]=j%2+1;  
  
    a[i]=dummy[DUMMY_LENGTH-2]*i;  
    b[i]=dummy[DUMMY_LENGTH-1]*i+1;  
}
```

быдло-  
код 😜



```
/PGP-2023> nvprof -m local_memory_overhead ./lab4 20 32
```

| Invocations                   | Metric Name           | Metric Description    | Min    | Max    | Avg    |
|-------------------------------|-----------------------|-----------------------|--------|--------|--------|
| Device "GeForce GTX 1050 (0)" |                       |                       |        |        |        |
| Kernel: gInit(int*, int*)     |                       |                       |        |        |        |
| 1                             | local_memory_overhead | Local Memory Overhead | 98.11% | 98.11% | 98.11% |
| Kernel: gSum(int*, int*)      |                       |                       |        |        |        |
| 1                             | local_memory_overhead | Local Memory Overhead | 0.00%  | 0.00%  | 0.00%  |

| Type            | Time (%) | Time            | Calls           | Min             | Max               |
|-----------------|----------|-----------------|-----------------|-----------------|-------------------|
|                 |          |                 | Avg             |                 | Name              |
| GPU activities: | 83.15%   | <b>4.3196ms</b> | 1               |                 |                   |
|                 |          |                 | <b>4.3196ms</b> | <b>4.3196ms</b> | <b>4.3196ms</b>   |
|                 |          |                 |                 |                 | gInit(int*, int*) |
|                 | 4.67%    | 242.60us        | 1               |                 |                   |
|                 |          |                 | <b>242.60us</b> | <b>242.60us</b> | <b>242.60us</b>   |
|                 |          |                 |                 |                 | gSum(int*, int*)  |

```
#define DUMMY_LENGTH 101
```

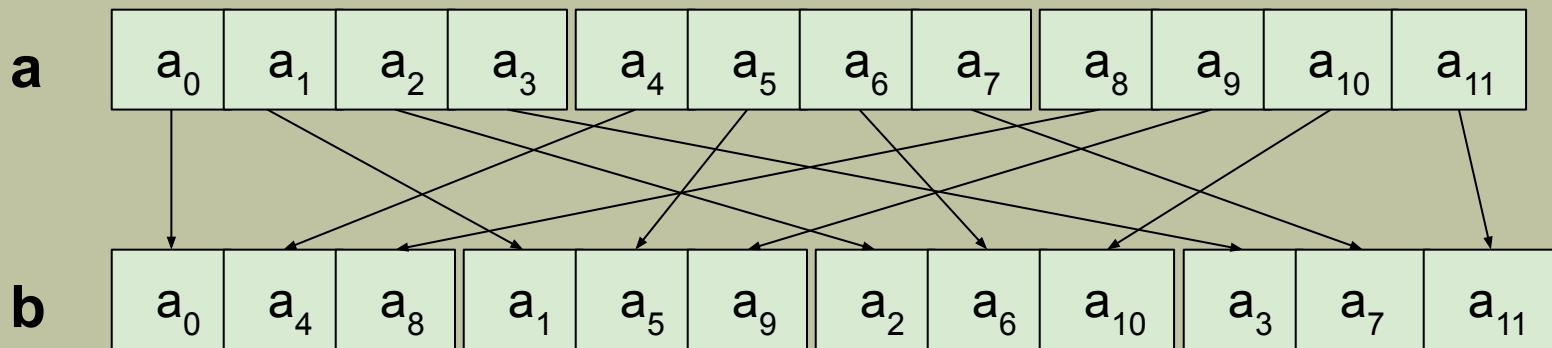
```
/PGP-2023> nvprof -m local_memory_overhead ./lab4 20 32
```

| Invocations                   | Metric Name           | Metric Description    | Min   | Max   | Avg   |
|-------------------------------|-----------------------|-----------------------|-------|-------|-------|
| Device "GeForce GTX 1050 (0)" |                       |                       |       |       |       |
| Kernel: gInit(int*, int*)     |                       |                       |       |       |       |
| 1                             | local_memory_overhead | Local Memory Overhead | 0.00% | 0.00% | 0.00% |
| Kernel: gSum(int*, int*)      |                       |                       |       |       |       |
| 1                             | local_memory_overhead | Local Memory Overhead | 0.00% | 0.00% | 0.00% |

| Type            | Time (%) | Time            | Calls           | Min               | Max             |
|-----------------|----------|-----------------|-----------------|-------------------|-----------------|
|                 |          |                 | Avg             | Name              |                 |
| GPU activities: | 16.67%   | <b>173.67us</b> | 1               |                   |                 |
|                 |          |                 | <b>173.67us</b> | <b>173.67us</b>   | <b>173.67us</b> |
|                 |          |                 |                 | gInit(int*, int*) |                 |
|                 | 23.26%   | <b>242.25us</b> | 1               |                   |                 |
|                 |          |                 | <b>242.25us</b> | <b>242.25us</b>   | <b>242.25us</b> |
|                 |          |                 |                 | gSum(int*, int*)  |                 |

## Лабораторная 5

Провести копирование массива  $a$ , включающего  $N$  векторов длины  $K$  по образцу, приведенному на диаграмме:



|       |       |          |          |
|-------|-------|----------|----------|
| $a_0$ | $a_1$ | $a_2$    | $a_3$    |
| $a_4$ | $a_5$ | $a_6$    | $a_7$    |
| $a_8$ | $a_9$ | $a_{10}$ | $a_{11}$ |

**Транспонирование**



|       |       |          |
|-------|-------|----------|
| $a_0$ | $a_4$ | $a_8$    |
| $a_1$ | $a_5$ | $a_9$    |
| $a_2$ | $a_6$ | $a_{10}$ |
| $a_3$ | $a_7$ | $a_{11}$ |

$n=i/K$ ,  $k=i\%N$  - двумерная индексация, матрица.  
 $a[k+n*K] \Rightarrow a[n+k*N]$